

Planning and MPC Equations

AC Track Control

1 Reference-line optimization

The lateral offset field o_i is optimized over a closed racing line while penalizing curvature, boundary violations, and non-smooth offsets:

$$J_{\text{line}} = t_{\text{lap}} + w_{\kappa} \sum_i \kappa_i^2 \Delta s_i + w_c \sum_i \max(0, c_{\text{target}} - c_i)^2 \Delta s_i + w_s \sum_i \|\Delta^2 o_i\|_2^2, \quad (1)$$

where t_{lap} is the static lap-time proxy, κ_i is the local curvature, c_i is the distance from the racing line to the nearest track boundary, and Δs_i is the segment length.

2 Speed envelope

The curvature-limited speed is

$$v_{\kappa,i} = \text{clamp}\left(\sqrt{\frac{a_{\text{lat}}}{|\kappa_i|}}, v_{\min}, v_{\max}\right). \quad (2)$$

The forward acceleration and backward braking passes enforce

$$v_i^2 \leq v_{i-1}^2 + 2a_x^{\text{accel}} \Delta s_{i-1}, \quad (3)$$

$$v_i^2 \leq v_{i+1}^2 + 2a_x^{\text{brake}} \Delta s_i. \quad (4)$$

The high-speed planning response time is

$$\tau_{\text{eff}} = \tau_0 + \max(0, v - 200) g_{\text{hs}}, \quad v \text{ in km/h}, \quad (5)$$

and the braking-limited speed satisfies

$$v_{\text{allowed}}^2 \leq v_{\text{ahead}}^2 + 2a_{\text{brake}} d_{\text{eff}}, \quad d_{\text{eff}} = d - v\tau_{\text{eff}}. \quad (6)$$

3 Lateral MPC

The Frenet state and control input are defined as

$$x = [e_y \quad e_{\psi} \quad v_x \quad v_y \quad r]^{\top}, \quad u = [\delta \quad a_x]^{\top}. \quad (7)$$

The speed-scheduled continuous-time model is

$$\dot{e}_y = v_x e_{\psi} + v_y, \quad (8)$$

$$\dot{e}_{\psi} = r, \quad (9)$$

$$\dot{v}_x = a_x, \quad (10)$$

$$\dot{v}_y = a_{33}(v_x)v_y + a_{34}(v_x)r + b_{31}(v_x)\delta, \quad (11)$$

$$\dot{r} = a_{43}(v_x)v_y + a_{44}(v_x)r + b_{41}(v_x)\delta. \quad (12)$$

The identified coefficients are scheduled by speed and augmented by the load-dependent yaw-response factor $\gamma(|a_y|)$:

$$a_{33}(v_x) = -\frac{c_{\text{lat}}(v_x)}{v_x}, \quad a_{34}(v_x) = \frac{c_{\text{yaw,lat}}(v_x)}{v_x} - v_x, \quad (13)$$

$$a_{43}(v_x) = \frac{c_{\text{yaw,vel}}(v_x)}{v_x}, \quad a_{44}(v_x) = \frac{c_{\text{yaw,damp}}(v_x)}{v_x}. \quad (14)$$

With the steering actuator state, the model is extended by

$$\dot{\delta} = \frac{\delta_{\text{cmd}} - \delta}{\tau_{\text{steer}}}. \quad (15)$$

For a horizon of N stages, the LMPC cost is

$$J_{\text{lat}} = \sum_{k=0}^{N-1} \left[(x_k - x_{\text{ref},k})^\top Q_k (x_k - x_{\text{ref},k}) + u_k^\top R_k u_k + \Delta u_k^\top R_{d,k} \Delta u_k \right] + (x_N - x_{\text{ref},N})^\top Q_N (x_N - x_{\text{ref},N}), \quad (16)$$

where

$$Q_k = \text{diag}(q_y m_y(v_k), q_\psi m_\psi(v_k), q_{v_x}, q_{v_y}, q_r m_r(v_k)), \quad (17)$$

and m_y , m_ψ , and m_r are speed-dependent weight maps.

4 Longitudinal MPC

The response lag is discretized by

$$\alpha = \frac{\Delta t}{\tau_{\text{response}}}, \quad 0 \leq \alpha \leq 1. \quad (18)$$

The incremental acceleration model is

$$a_{k+1} = (1 - \alpha)a_k + \alpha u_k, \quad (19)$$

$$v_{k+1} = v_k + \Delta t (a_{k+1} - a_{\text{coast}}(v_k)). \quad (20)$$

The longitudinal cost combines speed tracking, acceleration effort, and jerk:

$$J_{\text{lon}} = \sum_{k=0}^{N-1} \left[q_v (v_k - v_{\text{ref},k})^2 + q_a (a_k - a_{\text{ref},k})^2 + r_a \Delta a_k^2 + r_j (\Delta^2 a_k)^2 \right]. \quad (21)$$

The acceleration and jerk constraints are

$$-a_{\text{brake,max}} \leq a_k \leq a_{\text{accel,max}}, \quad (22)$$

$$-j_{\text{brake,max}} \Delta t \leq \Delta a_k \leq +j_{\text{accel,max}} \Delta t. \quad (23)$$

5 Pedal mapping

The acceleration request is converted to throttle by

$$T = \text{clamp}\left(\frac{a_{\text{propulsion}}}{a_{\text{cap}}(v)}, 0, 1\right), \quad (24)$$

and to brake by interpolating the measured deceleration grid:

$$B = \mathcal{I}_{v,p}(a_{\text{decel}}), \quad p \in \{0, 0.15, 0.30, 0.45, 0.60, 0.75, 0.90, 1.00\}. \quad (25)$$

6 Steady-state axis-response consistency relation

The normalized-axis response can be projected onto the steady-state relation

$$\frac{uv}{r} = \frac{L}{k_{\text{eff}}} + \frac{K_{\text{eff}}}{k_{\text{eff}}} v^2, \quad (26)$$

where L is the wheelbase. The effective scale k_{eff} and derived understeer gradient K_{eff} are not independently identifiable physical steering parameters unless the road-wheel angle is measured separately. They summarize the same normalized-axis response already identified by the dynamic model and are used here as a consistency check.

7 Linear model and steering-actuator augmentation

For the five-state Frenet model, the continuous-time state matrix is

$$A_c = \begin{bmatrix} 0 & v_x & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{43} & a_{44} \end{bmatrix}, \quad B_c = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \\ b_{31} & 0 \\ b_{41} & 0 \end{bmatrix}. \quad (27)$$

The speed-scheduled coefficients are

$$a_{33} = -\frac{c_{\text{lat}}}{v_x}, \quad a_{34} = \frac{c_{\text{yaw, lat}}}{v_x} - v_x, \quad (28)$$

$$a_{43} = \frac{c_{\text{yaw, vel}}}{v_x}, \quad a_{44} = \frac{c_{\text{yaw, damp}}}{v_x}. \quad (29)$$

With an actuator state δ , the augmented continuous model is

$$\begin{bmatrix} \dot{e}_y \\ \dot{e}_\psi \\ \dot{v}_x \\ \dot{v}_y \\ \dot{r} \\ \dot{\delta} \end{bmatrix} = A_{\text{aug}} \begin{bmatrix} e_y \\ e_\psi \\ v_x \\ v_y \\ r \\ \delta \end{bmatrix} + B_{\text{aug}} \begin{bmatrix} \delta_{\text{cmd}} \\ a_x \end{bmatrix}, \quad (30)$$

where

$$A_{\text{aug}} = \begin{bmatrix} 0 & v_x & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & a_{33} & a_{34} & b_{31} \\ 0 & 0 & 0 & a_{43} & a_{44} & b_{41} \\ 0 & 0 & 0 & 0 & 0 & -1/\tau_{\text{steer}} \end{bmatrix}, \quad (31)$$

$$B_{\text{aug}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \\ 1/\tau_{\text{steer}} & 0 \end{bmatrix}. \quad (32)$$

The sampled model used by the QP is

$$A_d = I + A_{\text{aug}}\Delta t, \quad B_d = B_{\text{aug}}\Delta t, \quad (33)$$

with the discrete spectral radius bounded before QP assembly.